

BUILDING A UNIVERSITY TIMETABLE TAKES

~~Weeks.~~

Seconds.

Come see UMTAS live: smarter timetables and campus tools for students, lecturers and admins.

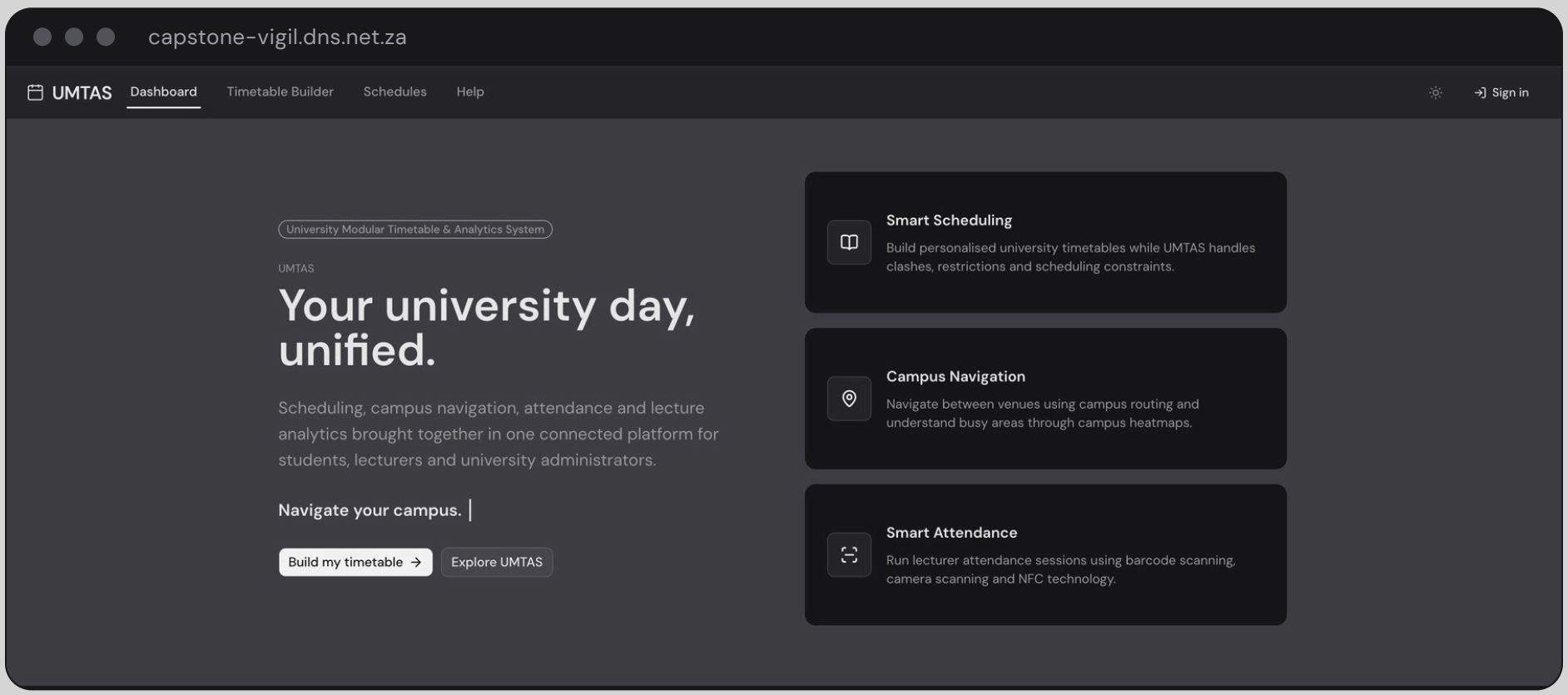
UMTAS

Fri 30 Oct · Rautenbach Hall, UP Hatfield

Swipe →

One platform for campus life.

UMTAS, the University Modular Timetable & Analytics System, solves university timetables and helps students, lecturers and administrators manage campus life.



Students

A clash-free week and a route to every class.

Lecturers

Attendance in seconds and a read on the room.

Admins

Timetables, calendars and campus crowds in one place.

Live now at capstone-vigil.dns.net.za

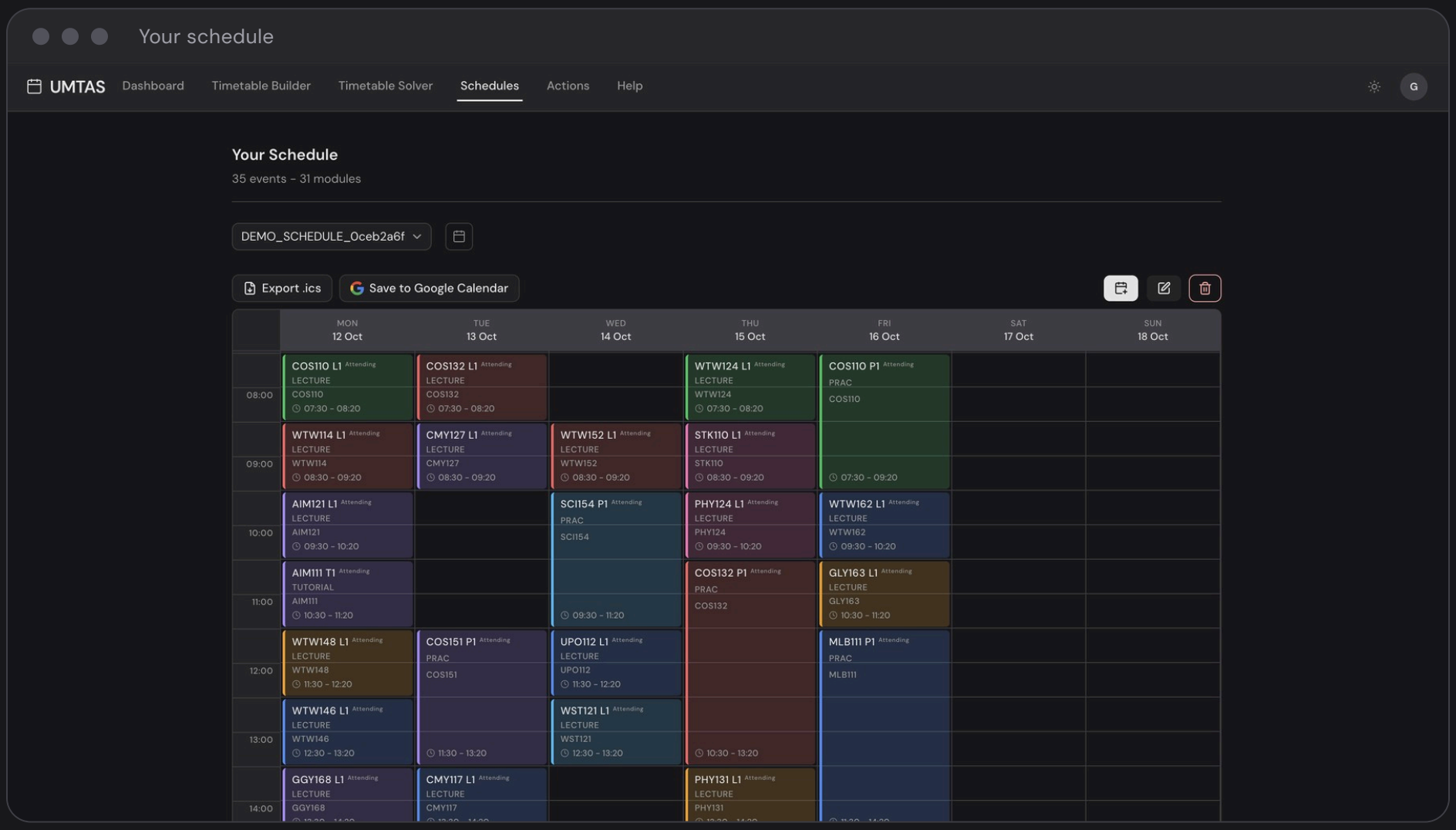
Swipe →

FEATURE 01

INTELLIGENT TIMETABLE GENERATION

Timetables, solved.

Upload the official timetable PDF, set your preferences, and get a week with no clashes.



Reads your PDF
Modules and events, imported without retyping.

Your preferences
Later starts, a free day, fewer gaps.

Synced
Google Calendar and ICS export.

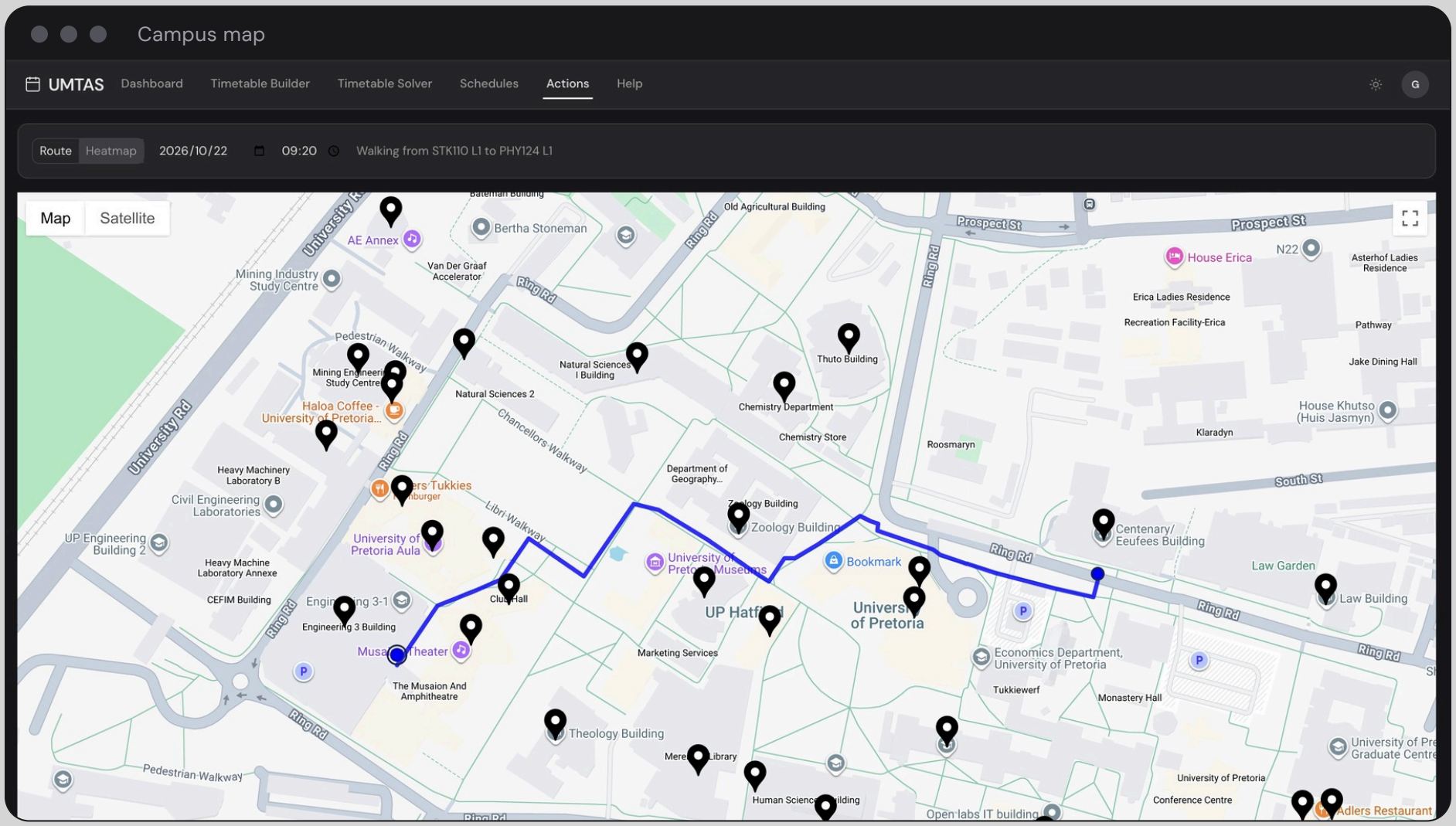
UMTAS

Swipe →

CAMPUS MAPS, ROUTING & HEATMAPS

Find your way. Skip the crowds.

Walking routes between your classes, and heat maps that show where campus gets busy.



Routes

Directions from one venue to the next.

Heatmaps

Projected foot traffic, hour by hour.

Rerouting

Admins can send students along quieter paths.

UMTAS

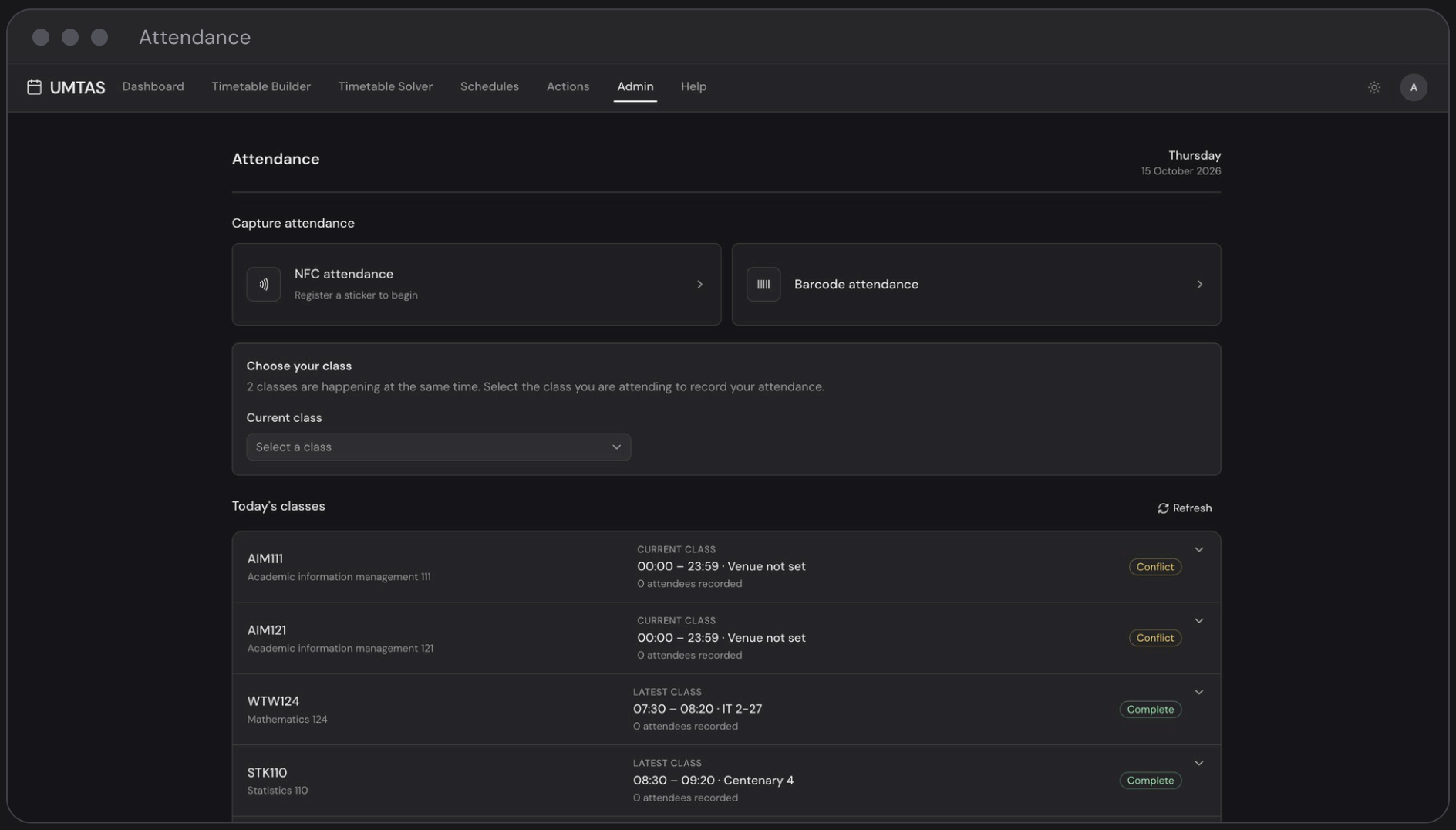
Swipe →

FEATURE 03

ATTENDANCE TRACKING

Roll call in seconds.

Students check themselves in. Lecturers see who's there, without passing a register around.



Barcode

Scan student cards with a USB scanner.

Camera

Or scan cards with a laptop camera.

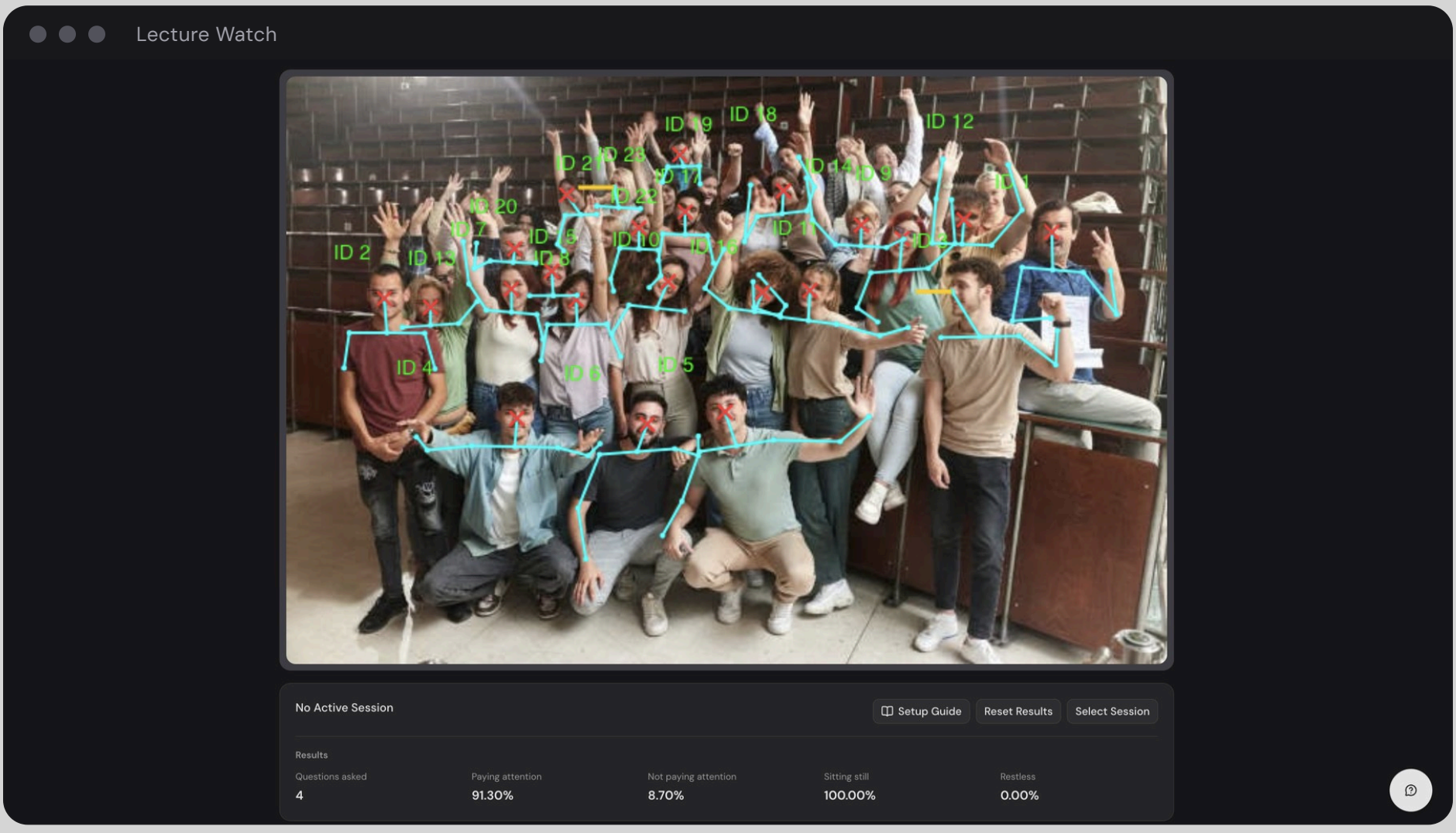
NFC

Students tap the lecturer's sticker with their phone.

LECTURE WATCH

Read the room.

A vision model gives lecturers class-level engagement insights, right in the browser.



Engagement

Raised hands, attention and restlessness.

On-device

Video is analysed in the browser.

Private

Only class-level numbers are saved, never footage.

Built like production software.

A real system that real people log into, tested and monitored like one.

1,100+

automated tests: unit, integration, end-to-end, parser and solver

17/17

quality requirements pass, from speed to security to accessibility

99.97%

public uptime over 30 days

640 ms

95th-percentile response across 40,513 requests at 100 users at once

2 engines

Google OR-Tools CP-SAT, with an evolutionary fallback

3,606

commits in 25 weeks, reviewed through pull requests

Next.js · NestJS · PostgreSQL · Redis · FastAPI · Docker · Grafana

Swipe →

Meet Team Vigil.



Wilmar Smit

TEAM LEAD & INTEGRATION LEAD



Michael Tomlinson

LEAD DEVELOPER & SYSTEM ARCHITECT



Johan Coetzer

FRONTEND LEAD & FULL-STACK DEVELOPER



Marcel Stoltz

DEVOPS LEAD & BACKEND SPECIALIST



Aidan Dawson

BACKEND DEVELOPER & INTEGRATION

WITH THANKS TO OUR INDUSTRY PARTNERS

Tyto Insights & DNS Business

Mentored by Inge Odendaal & Vincent Feistel

SEE IT LIVE

Come see UMTAS live.

Friday, 30 October 2026 · 09:00–13:00

Rautenbach Hall, UP Hatfield Campus

RSVP

Let us know you're coming

Try it now

capstone-vigil.dns.net.za

Star the repo

github.com/COS301-SE-2026/UMTAS

UMTAS

Team Vigil · University of Pretoria
COS 301 Capstone 2026